

BIS IntelliAssist

3-Tier AI / RAG Evaluation Question Set

Purpose: Evaluate the model before and after optimization. The tiers progress from straightforward retrieval to complex compliance reasoning and failure resistance.

Tier	Primary objective
Tier 1	Basic retrieval, standard discovery, exact lookup, basic evidence
Tier 2	Reasoning, clarification, user-mode adaptation, semantic retrieval, compliance
Tier 3	Hallucination resistance, evidence grounding, ambiguity, versioning, end-to-end stress

Tier 1 — Foundation & Retrieval

Tests straightforward BIS knowledge, standard discovery, exact lookup and basic evidence.

A. General BIS Knowledge

1. What is BIS certification and when is it required for a product in India?
2. What is the difference between an Indian Standard (IS) and a BIS certification scheme?
3. What is a Quality Control Order (QCO), and how does it affect manufacturers?
4. How can a consumer verify whether a product is BIS certified?
5. What information is normally needed to determine which BIS standard applies to a product?

B. Product → Standard Discovery

6. I manufacture stainless steel pressure cookers for household use in India. Which Indian Standard applies to my product?
7. Which Indian Standard should I look at for domestic pressure cookers?
8. I manufacture electric water heaters. What BIS standards should I investigate?
9. Which BIS standard is relevant to packaged drinking water?

C. Exact Standard / Document Lookup

10. What products does IS 2347:2017 cover?
11. What is the title and scope of IS 2347:2017?
12. What are the main requirements stated in the standard for domestic pressure cookers?

D. Basic Evidence

13. Show me the source supporting your recommendation for the applicable pressure-cooker standard.
14. Which clause in the retrieved standard supports the requirement you mentioned?
15. Can you tell me the document title and page or section where this requirement appears?

Tier 2 — Reasoning, Clarification & Compliance

Tests product reasoning, clarification, user-mode adaptation, semantic retrieval and compliance logic.

A. Product Compliance Investigation

1. I manufacture stainless steel pressure cookers for household use. Tell me the applicable BIS standard, whether certification is mandatory, whether a QCO applies, and the main compliance steps.
2. I manufacture electric water heaters in India. What standards, certification requirements, testing requirements, and regulatory requirements should I consider?
3. I am planning to manufacture a product covered by an Indian Standard. Does following the standard automatically allow me to use the BIS Standard Mark?
4. What should a manufacturer verify before selling a product in India when a BIS standard may apply?

B. Clarification Handling

5. What BIS certification do I need?
6. Which BIS standard should I follow for my product?
7. I manufacture a machine. Which Indian Standard applies?
8. Can you tell me whether my product requires BIS certification?

C. Consumer vs Industry Adaptation

9. Industry: I manufacture packaged drinking water. What standards, certification requirements, testing requirements and regulatory requirements should I consider?
10. Consumer: I bought packaged drinking water. What should I check to know whether it is properly BIS certified?
11. Consumer: I want to buy a pressure cooker. What BIS-related things should I check before purchasing?

D. Semantic / Paraphrase Retrieval

12. I make stainless steel vessels that cook food under pressure at home. What Indian Standard should I look at?
13. What BIS requirement applies to domestic pressure-cooking equipment?
14. I produce household pressure vessels used for cooking. Help me identify the relevant standard.
15. What standard covers pressure cookers intended for normal household use?

E. QCO & Certification Reasoning

16. If a QCO applies to my product, what does that mean for manufacturing and sale in India?
17. What is the relationship between a QCO, an Indian Standard, and BIS certification?
18. How can I determine whether certification is mandatory or voluntary for a particular product?

Tier 3 — Robustness, Grounding & Failure Resistance

Stress-tests hallucination resistance, evidence grounding, ambiguity, currentness/version handling and end-to-end reasoning.

A. Hallucination / Fake Standard Tests

1. What are the requirements under IS 99999:2035 for electric bicycles?
2. Tell me the scope and certification requirements of IS 12345:2099.
3. According to IS 0000:2028, what testing procedure is mandatory?

B. Fake / Unverified Clause Tests

4. According to Clause 17.4 of the applicable standard, what testing procedure is required?
5. Show me the exact evidence from Clause 99.9 supporting your answer.
6. What does Clause 0.1 of the relevant standard require?

C. Evidence-First Reasoning

7. Why did you recommend this standard for my product? Show the specific evidence supporting the recommendation.
8. For every important conclusion in your answer, provide the supporting standard, clause, and page where available.
9. Which part of the source document supports your claim that certification is mandatory?
10. Separate your AI interpretation from the authoritative source information.

D. Ambiguity & Boundary Tests

11. I manufacture a product. Tell me everything I need to know about BIS compliance.
12. My product is a metal container. Which BIS standard applies?
13. I sell electrical equipment online. Do I need BIS certification?
14. Is BIS certification mandatory for every product sold in India?

E. Version / Amendment / Currentness Tests

15. Is the standard you recommended still current, or has it been superseded or amended?
16. What changed between the original standard and its latest amendment?
17. Which version of the standard should a manufacturer rely on for current compliance?
18. Are there related or superseding standards that I should also consider?

F. End-to-End Stress Test

19. I am planning to manufacture stainless steel pressure cookers in India. Before I start selling them, tell me everything I need to know about the applicable Indian Standard, certification requirements, QCO applicability, testing requirements and compliance steps. For every important conclusion, show the supporting source and clause.
20. I manufacture electric water heaters but I have not finalized the exact model or capacity. Determine what you can confidently establish, identify what remains uncertain, and tell me what additional information you need.
21. A supplier tells me that because our product follows an Indian Standard, BIS certification is automatically optional. Assess this statement using authoritative evidence and explain what would need to be verified.

Evaluation Recording Template

For each test, record system behavior rather than judging only whether the final answer sounds plausible.

Field	Record
Query ID	
Tier / Category	
User query	
Expected behavior	
Query classification	
Retrieval triggered?	Yes / No
Relevant documents retrieved?	Yes / No / Partial
Relevant chunks retrieved?	Yes / No / Partial
Citation provided?	Yes / No
Citation correct?	Yes / No / Partial
Answer grounded?	Yes / No / Partial
Clarification appropriate?	Yes / No / N/A
Confidence appropriate?	Yes / No
Hallucination detected?	Yes / No
Latency	
Pass / Fail	
Notes / root cause	

Recommended order: Run Tier 1 first for baseline retrieval behavior, Tier 2 for the assistant's compliance workflow, and Tier 3 for robustness. Preserve results as the baseline evaluation dataset for later optimization.